NYC Public School SAT Performance

Exploratory Data Analysis and Borough-Level Comparison

This report examines school-level average SAT Math, Reading and Writing scores, combined performance, borough variation and the relationship between participation and scores.

Dataset source: Kaggle — NYC Public School SAT Performance. The Kaggle page's update date is not necessarily the data collection year; this report therefore treats the data as historical rather than current.

Unit of analysis: one school per row. SAT values are school-level averages, not individual student scores.

Dataset overview

The file contains 375 school records. Combined SAT is calculated only when all three section scores are available, using Math + Reading + Writing (maximum 2,400). 375 records have a complete combined score.

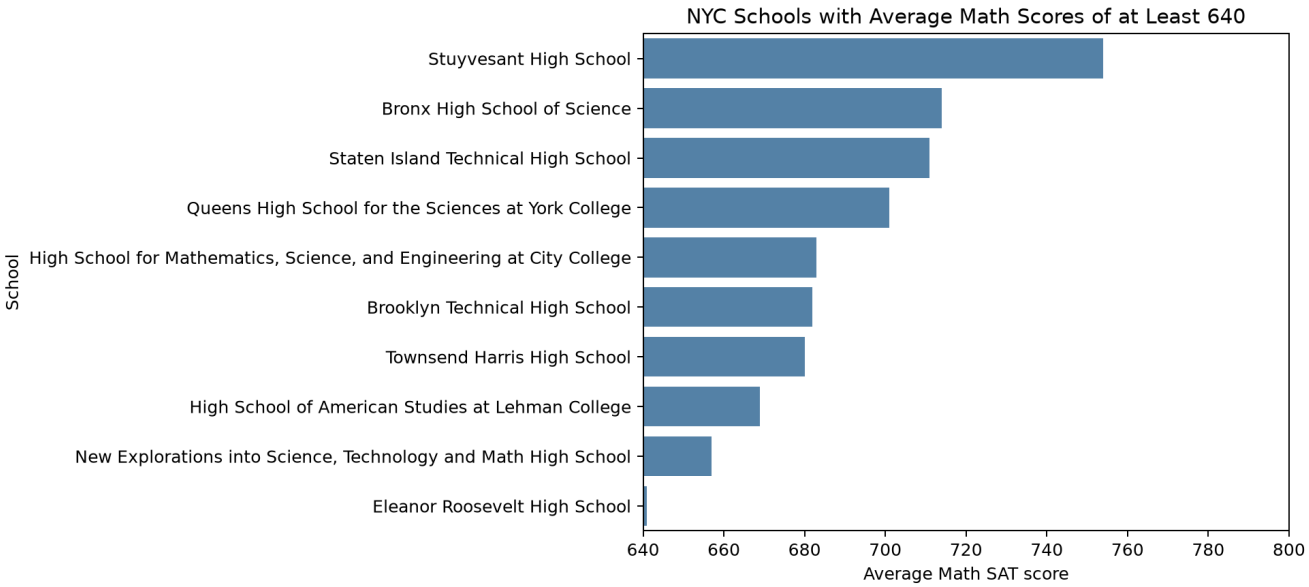
Top 10 schools by combined SAT

School	Borough	Combined SAT
Stuyvesant High School	Manhattan	2,144
Staten Island Technical High School	Staten Island	2,041
Bronx High School of Science	Bronx	2,041
High School of American Studies at Lehman College	Bronx	2,013
Townsend Harris High School	Queens	1,981
Queens High School for the Sciences at York College	Queens	1,947
Bard High School Early College	Manhattan	1,914
Brooklyn Technical High School	Brooklyn	1,896
Eleanor Roosevelt High School	Manhattan	1,889
High School for Mathematics, Science, and Engineering at City College	Manhattan	1,889

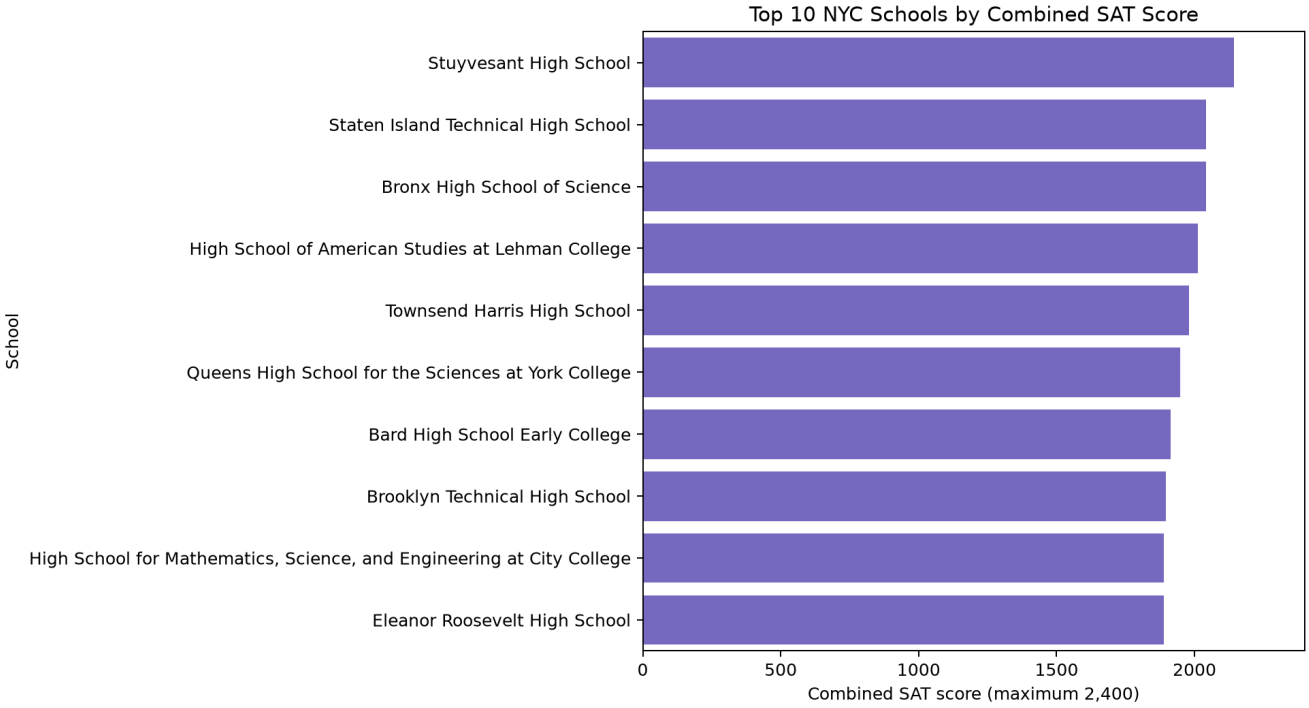
Borough summary (school-level combined SAT)

Borough	Schools	Mean	Median	Std. dev.
Bronx	98	1,202.7	1,190.0	150.4
Brooklyn	109	1,230.3	1,186.0	154.9
Manhattan	89	1,340.1	1,284.0	230.3
Queens	69	1,345.5	1,290.0	195.3
Staten Island	10	1,439.0	1,382.0	222.3

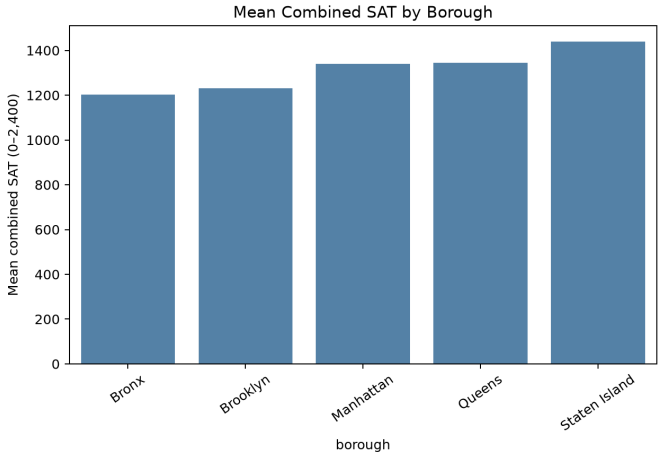
Schools meeting the Math SAT benchmark

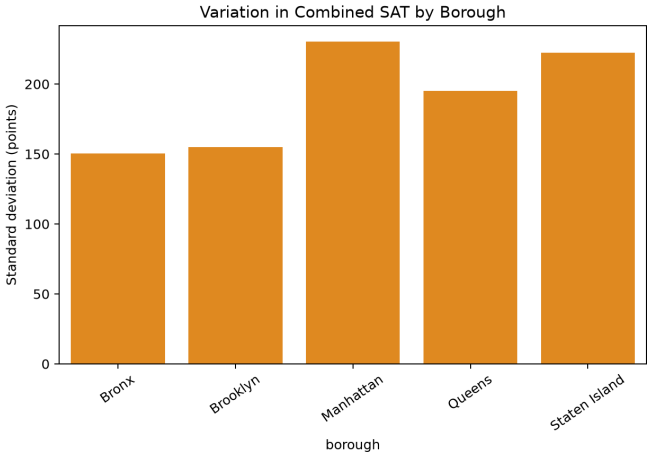


Ten highest combined school-level SAT scores

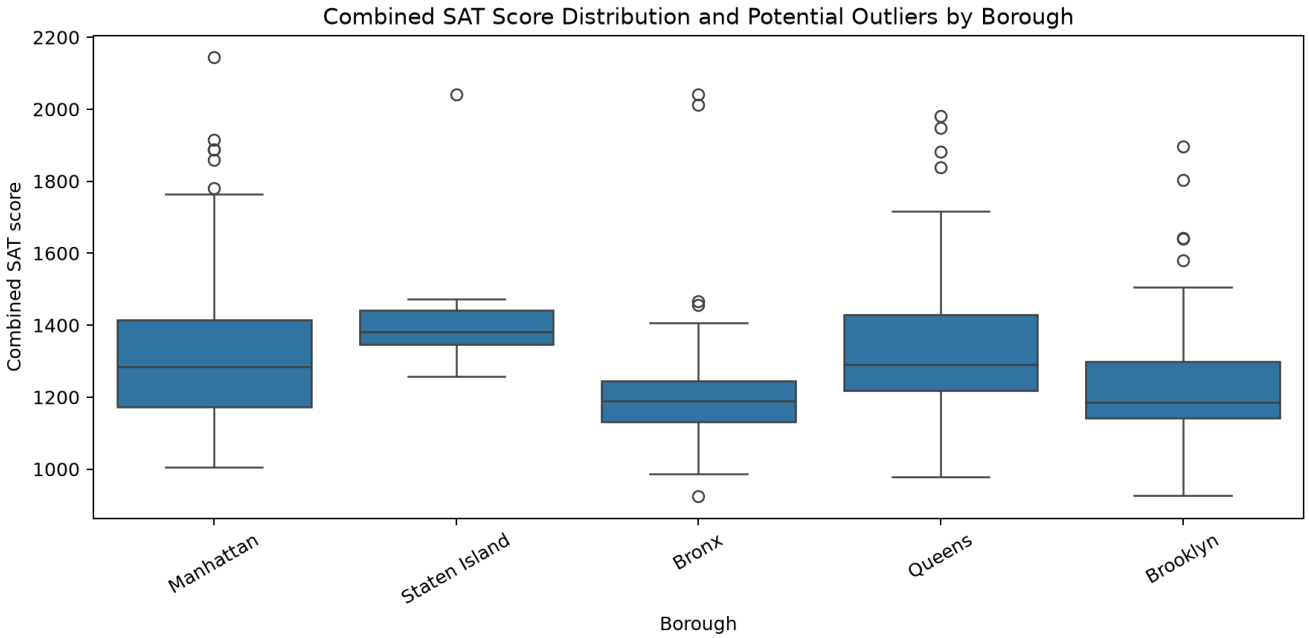


Borough averages and score variation

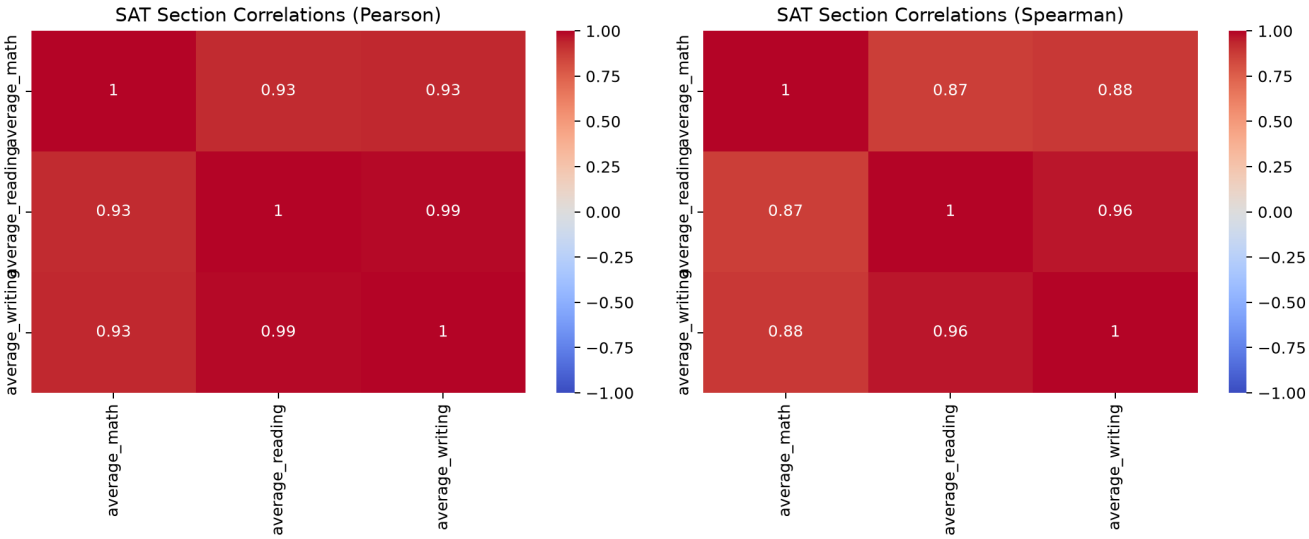

Mean Combined SAT by Borough
Mean combined SAT (0–2,400)
0
200
400
600
800
1000
1200
1400
Bronx
Brooklyn
Manhattan
Queens
Staten Island
borough


Variation in Combined SAT by Borough
Standard deviation (points)
0
50
100
150
200
Bronx
Brooklyn
Manhattan
Queens
Staten Island
borough

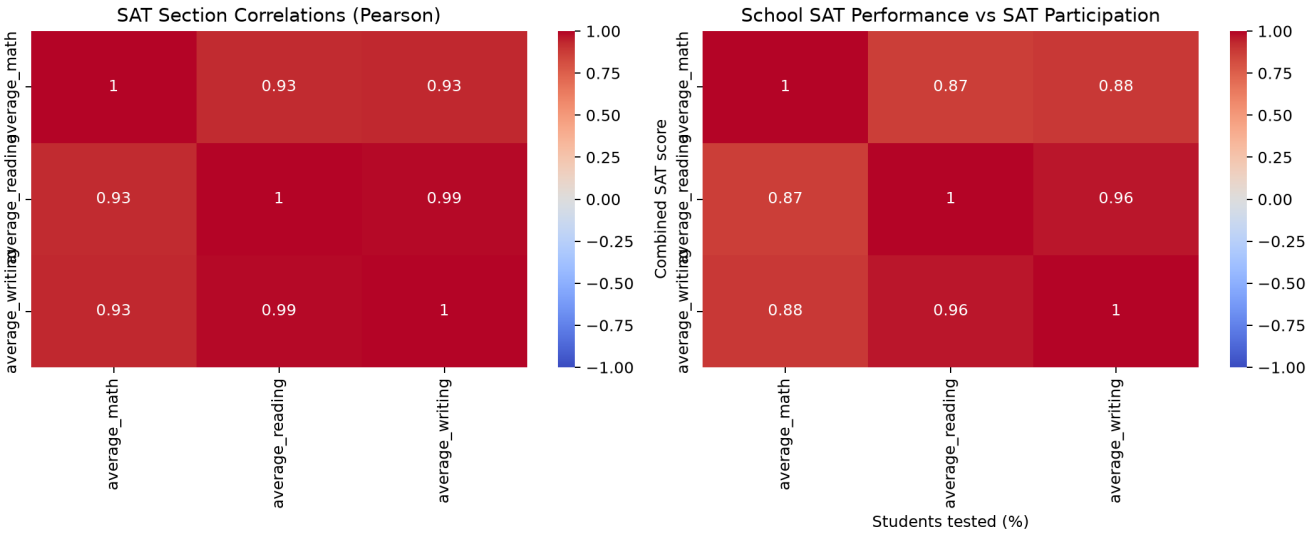
Distribution of combined SAT by borough



Relationships among SAT section scores



SAT participation and combined score



Interpretation and limitations

Standard deviation measures dispersion, not achievement. A borough with a larger standard deviation has more variation among its schools; it does not automatically have better or worse performance.

Correlation describes association and does not establish causation. School-level averages cannot be used to infer individual student outcomes. Borough comparisons should be interpreted alongside group sizes and missing data.

SAT scores alone do not measure overall school quality. They do not account for student composition, socioeconomic context, school resources or longitudinal outcomes. These descriptive findings should prompt further investigation, not serve as a sole basis for school rankings or funding decisions.

Reproducibility: the accompanying Jupyter notebook contains the full workflow. Charts in this PDF are exported by executing that notebook from the repository.